Solvency Field Theory

V10.9 Companion Note

Theorem 6b: Strong-Field Gate, Vacuum No-Hair Pretests, and Branch Completion Status

Status: Lead 6b branch preserved through background, perturbative, and strong-field pretests. Exact static spherical closure remains open.

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1. Purpose of this note

V10.7 selected the lead minisuperspace branch of Theorem 6b: a **single-field, slaved/constrained, count-normalized branch** in which the bookkeeping scalar is sourced by the operation-density current and the count sector supplies the final $4\pi^2$ normalization. V10.8 then showed that this same branch is **structurally compatible with the V10.1 perturbative source sector**.

The main unresolved danger point after V10.8 was the **strong-field gate**. The question was simple and load-bearing:

> **Does the lead single-field branch develop unacceptable vacuum scalar hair outside compact sources, or does it decouple cleanly enough that Schwarzschild/GR recovery remains plausible?**

This note records the first two strong-field pretests of that question. The conclusion is not that exact Schwarzschild has been derived. The conclusion is narrower and more useful:

> **The lead large-χ vacuum branch passes a strong no-hair pretest on both flat-radial and fixed Schwarzschild backgrounds. There is no obvious vacuum scalar-hair obstruction in the physically relevant complete-screening branch.**

That moves Theorem 6b out of branch-selection mode and into final-closure mode.

2. What V10.7 and V10.8 already established

The lead 6b branch is the single-field constrained/slaved action

$$S_{1\mathrm{f}} = \int d^4x\,\sqrt{-g}\,\Big[\frac{R}{16\pi G} - \frac{K}{2}\nabla_\mu\chi\nabla^\mu\chi - V_0 e^{-\chi} - \eta\,\nabla_\mu\chi\,J^\mu_{\mathrm{ops}} + \mathcal{L}_m + \Lambda_C\,\mathcal{C}[g,\chi,\psi]\Big],$$

with

$$u^\mu = \frac{\nabla^\mu \chi}{\sqrt{-\nabla_\alpha \chi \nabla^\alpha \chi}}, \qquad h^{\mu\nu} = g^{\mu\nu} + \nu^\mu \nu^\nu, \qquad J^\mu_{\rm ops} = \rho_{\rm ops}\, \nu^\mu,$$

and

$$\rho_{\rm ops} = \frac{T_{\mu\nu}\nu^\mu\nu^\nu + h^{\mu\nu}T_{\mu\nu}}{h} = \frac{\rho + 3p}{h} \quad \text{in the local clock-field frame.}$$

The sourced background equation is

$$K(\ddot{\chi} + 3H\dot{\chi}) - V_0 e^{-\chi} = \eta\, \nabla_\mu J^\mu_{\rm ops}, \qquad \nabla_\mu J^\mu_{\rm ops} = -\frac{3Hw(1+3w)\rho}{h}.$$

So dust gives zero driving while radiation gives active driving. In the slaved limit and with the count multiplier

$$\beta = \frac{1}{4\pi^2},$$

the background branch gives dust exactly and radiation asymptotically exactly. V10.8 then showed that the same branch is structurally compatible with the V10.1 perturbative source sector:

$$S_{\rm eff}(k,\tau) \sim \beta_Q^{\rm eff}(\tau)\, \frac{k^2}{a(\tau)^2}\, \Phi_{\rm eq}^{\rm eff}(\tau)\, W\!\left(\frac{k}{k_c^{\rm eff}(\tau)}\right),$$

with the same phase-curvature operator, the same instability-window logic, and finite-epoch source kernels.

So the remaining question is now specifically strong-field: does the vacuum exterior decouple?

3. Why the strong-field gate matters

The source coupling vanishes in vacuum because

$$J^\mu_{\rm ops} = 0 \qquad \text{outside ordinary matter and radiation.}$$

The maintenance potential also vanishes on the complete-screening branch because

$$V(\chi) = V_0 e^{-\chi} = V_0(1-F) \longrightarrow 0 \quad \text{as} \quad \chi \to \infty,$$

which is exactly the 6a screening/completion logic carried into the action.

So the structural expectation has always been that the strong-field branch should recover pure GR in vacuum. But until one checks the static spherical equations, that remains a structural argument, not a demonstrated result.

The right first question is therefore not "is exact Schwarzschild already proven?" The right first question is:

> **Does the vacuum branch want to carry significant exterior scalar hair, or does that hair collapse in the physically relevant large-χ branch?**

4. Strong-field pretest 1: flat-radial vacuum exterior

The simplest static spherical vacuum reduction is

$$K\left(\chi'' + \frac{2}{r}\chi'\right) = V_0 e^{-\chi},$$

with asymptotic boundary conditions

$$\chi(R_{\max}) = \chi_\infty, \qquad \chi'(R_{\max}) = 0.$$

The exterior scalar-energy proxy is

$$\rho_\chi(r) = \frac{K}{2}\chi'(r)^2 + V_0 e^{-\chi(r)},$$

and the integrated exterior scalar mass proxy is

$$M_\chi = 4\pi \int_{R_s}^{R_{\max}} r^2 \rho_\chi(r)\, dr.$$

This pretest showed two distinct vacuum branches:

1. a **bad low-χ_∞ branch**, where scalar hair remains large,
2. a **good large-χ_∞ branch**, where scalar hair collapses rapidly.

For example, at small K and increasing χ_∞:

$$K = 10^{-4}, \quad \chi_\infty = 20: \qquad M_\chi \sim 5\times 10^{-4}, \quad \rho_\chi(R_s) \sim 6\times 10^{-6},$$

while

$$K = 10^{-4}, \quad \chi_\infty = 25: \qquad M_\chi \sim 3\times 10^{-6}, \quad \rho_\chi(R_s) \sim 3\times 10^{-10}.$$

So the good branch already decouples strongly in the simplest vacuum reduction.

5. Strong-field pretest 2: fixed Schwarzschild background

The more serious exterior pretest places the scalar on a fixed Schwarzschild background

$$ds^2 = -f(r)dt^2 + f(r)^{-1}dr^2 + r^2 d\Omega^2, \qquad f(r) = 1 - \frac{2M}{r},$$

and solves

$$K\,\frac{1}{r^2}\frac{d}{dr}\left(r^2 f(r)\frac{d\chi}{dr}\right) = V_0 e^{-\chi}.$$

The fixed-background scalar-energy proxy becomes

$$\rho_\chi(r) = \frac{K}{2} f(r)\chi'(r)^2 + V_0 e^{-\chi(r)}.$$

This matters because it asks the right question: does real exterior curvature itself revive a problematic scalar profile? If the good branch still decouples on Schwarzschild, then the strong-field gate becomes much less scary.

6. Robustness results on the Schwarzschild background

The robustness scan covered:

- masses $M = 0.1, 0.2, 0.3$,
- stiffnesses $K = 10^{-2}, 10^{-3}, 10^{-4}$,
- asymptotic branches $\chi_\infty = 20, 25$.

The result is very clear.

6.1. The $\chi_\infty = 25$ branch is excellent across the full scan

Across all scanned M and K values, the $\chi_\infty = 25$ branch remains in the excellent regime:

$$M_\chi \approx 9.26 \times 10^{-7} \quad \text{nearly invariant across the scan,}$$

with

$$\frac{M_\chi}{M} \sim 10^{-6}\text{–}10^{-5}, \qquad \rho_\chi(R_s) \ll 1, \qquad \max|\chi'| \ll 1 \quad \text{in the best cases.}$$

For the best decoupling case in the scan,

$$M = 0.3, \quad K = 10^{-2}, \quad \chi_\infty = 25,$$

one finds

$$M_\chi \approx 9.26 \times 10^{-7}, \qquad \frac{M_\chi}{M} \approx 3.09 \times 10^{-6}, \qquad \rho_\chi(R_s) \approx 6.05 \times 10^{-11}.$$

That is genuine vacuum decoupling.

6.2. The $\chi_\infty = 20$ branch is mostly good

The $\chi_\infty = 20$ branch remains good across almost the entire scan. The only corner that degrades to marginal is the smallest-mass, smallest-K case:

$$M = 0.1, \quad K = 10^{-4}, \quad \chi_\infty = 20.$$

Even there the branch is not catastrophically hairy; it is merely no longer in the clean decoupling regime. For larger M at the same K, the branch remains good.

So the physically relevant conclusion is:

The good large-χ vacuum branch survives the obvious Schwarzschild-background robustness test.

7. Interpretation of the branch structure

The strong-field pretests sharpen the vacuum structure of the lead 6b branch.

There are now clearly two vacuum branches:

1. a **bad lower-χ_∞ branch**, in which significant vacuum scalar hair can survive,
2. a **good large-χ_∞ branch**, corresponding to the physically relevant complete-screening branch, in which scalar hair collapses.

This is not an arbitrary mathematical curiosity. It is exactly what the maintenance/screening logic would suggest: the physically relevant vacuum branch in the strong field is the complete-screening branch, and that is precisely the branch in which the maintenance potential vanishes and scalar hair decouples.

So the strong-field gate is no longer "maybe the branch generically carries vacuum hair." It is now much narrower:

Does the full self-consistent static spherical solve select the same large-χ branch and recover exact Schwarzschild externally?

That is a much better problem.

8. What this does and does not establish

These pretests establish something real:

1. there is no obvious vacuum no-hair obstruction in the lead 6b branch,
2. the large-χ branch decouples strongly in vacuum,
3. that decoupling survives the first curved-background robustness scan.

But these pretests do **not** yet establish the full strong-field theorem. In particular they do not yet provide:

1. a full self-consistent metric-plus-scalar static spherical solution,
2. interior/exterior matching for compact sources,
3. exact Schwarzschild or exact TOV recovery derived from the coupled field equations.

So the correct status statement is not "strong-field gate closed." The correct status statement is:

The strong-field gate is now promising rather than scary.

9. Updated 6b completion status

After V10.9, the lead 6b branch has now passed or nearly passed all major selection gates:

- **Background gate:** passed by the single-field constrained/slaved branch, with dust exact and radiation asymptotically exact under count normalization.
- **Perturbative gate:** structurally passed, with correct phase-curvature operator, meaningful instability window, and finite-epoch source kernels compatible with V10.1 packaging.
- **Strong-field pretest gate:** passed on the large-χ branch, with no obvious vacuum scalar-hair obstruction in flat-radial or fixed Schwarzschild-background tests.

The branch-selection problem is therefore essentially over. The remaining work is now final formal closure:

1. the exact self-consistent static spherical solve,
2. the exact perturbative normalization translation into V10.1 language,
3. the fully covariant local count-constraint formulation.

10. Bottom line

The strongest current statement is:

> **The best live Theorem 6b branch is the single-field, slaved/constrained, count-normalized branch. In this branch dust is exact, radiation is asymptotically exact, the perturbative gate is structurally passed, and the strong-field vacuum branch passes a no-hair pretest on both flat-radial and Schwarzschild backgrounds.**

That does not yet close 6b. But it means the remaining work is no longer branch hunting or conceptual wilderness. It is final derivation work.